## ToolPool Error-Handling Rules:

**To get set up:**

git checkout main
git pull
git checkout -b mtunelife-milestone3

**What's on main now** (all in `backend-api/src/main/java/com/csc340/toolpool/`):

- `entity/User.java`, the User entity (matches the UML)
- `dto/UserRegistrationRequest.java` and `dto/UserResponse.java`
- `repository/UserRepository.java`, `service/UserService.java`, `controller/UserController.java` (`POST /api/users`)
- Shared error handling: `service/NotFoundException.java`, `service/BadRequestException.java`, `service/ConflictException.java`, and `controller/ApiExceptionHandler.java`

**Error handling rule (please follow this so our API is consistent):**

- Don't put `@ResponseStatus` on exceptions, and don't build error responses in controllers.
- In your **service**, just throw one of these:
  - `throw new NotFoundException("Tool 5 was not found.");` returns **404**
  - `throw new BadRequestException("dailyPrice must be greater than 0.");` returns **400**
  - `throw new ConflictException("This tool has rentals; unlist it instead.");` returns **409**
- `ApiExceptionHandler` automatically turns them into `{ "error": "your message" }`.
- `@Valid` failures on request bodies return 400 automatically, so you don't need to handle those.

**Patterns we're following (same as the demo):**

- Package by layer: `entity/`, `dto/`, `repository/`, `service/`, `controller/`
- Use request/response DTOs (records) like `UserRegistrationRequest` and `UserResponse`, not entities, in controllers

- Link objects by id in request bodies, like `"ownerId": 1`. Look the owner up in your service with `userRepository.findById(...)`, and throw `NotFoundException` if it's missing
- In `Tool.java`, the owner is `@ManyToOne(fetch = FetchType.LAZY, optional = false)` with `@JoinColumn(name = "owner_id", nullable = false)`. Don't add a list of tools to `User.java`; it isn't needed and avoids JSON recursion
- Put `@Service` and `@Transactional` on the service class
- Field names must match `docs/uml-class.png`

**When you're done:**

- Commit after each component (entity, repository, service, controller)
- Add your endpoints to the table in `backend-api/README.md` with "Implemented by" filled in, plus an example request body
- Open a PR to main with `Closes #41` in the description

For testing, I already created a test user (`chris@example.com`) on my Neon dev branch, but you'll be on your own dev branch, so POST a user first to use as the owner.